

Problem 11: New Tournament

10 Points

Problem ID: newtournament

Rank: 3

Introduction

After Big Big Big Big Big Ben defeated the evil [Tralalero Tralala](#), he becomes distraught without the [thrill of the fight](#) to keep him going. He tries getting a [j*b](#), but it's not the same as fighting. Spending day after day battling reports instead of warriors, Big Big Big Big Big Ben has reached [rock bottom](#). He realizes he must find a way to live again. He must find his [new tournament](#).

Problem Statement

Let N be the number of players in the tournament. N is always a power of 2.

Player X initially has power X at the beginning of the tournament. Each round, every consecutive pair of players plays each other; e.g. if the ordering of players is 1 2 3 4 5 6 etc., Player 1 plays Player 2, Player 3 plays Player 4, Player 5 plays Player 6, etc. The players may not necessarily start in consecutive order; *i.e.* the initial ordering is random. The winner is the player with the higher power, and the loser is deleted from the array. However, the winner's power then becomes the loser's power. The rounds continues until there's only one player remaining.

The players want to experiment with different starting positions. You are given their initial starting positions, but then also a shuffling pattern P that they will change positions by. Each shuffle iteration, the players' starting positions will be shuffled by the pattern P so that the person at position i ends up at position P_i afterwards. You are given Q queries you must answer. Each query contains a single number K . You must output the winner of the tournament if the starting position is shuffled by K iterations. The shuffled orders persist after queries.

Please note that both the initial position and the shuffle pattern are guaranteed to be generated randomly.

Input Format

The first line of the input contains a single integer T denoting the number of test cases that follow. For each test case:

- The first line contains two space separated integers N and Q , representing the number of players and the number of queries respectively.
- The next line contains N integers representing the initial starting positions of each player.
- The next line contains N integers representing the shuffling pattern P that the players will rearrange themselves by each shuffle iteration.
- The remaining Q lines contain an integer K representing the number of shuffle iterations in each query.

Output Format

For each query, you must respond with the winner of the tournament after K iterations of rotations.

Constraints

$$1 \leq T \leq 10$$

$$1 \leq N \leq 32768$$

$$1 \leq Q \leq 10^4$$

$$1 \leq K \leq 10^5$$

It is guaranteed that the sum of N across all test cases is less than 32768 and the sum of all Q across all test cases is less than 10^4 .

It is guaranteed that the initial position and rotation permutation for each test case are generated randomly.

Sample Test Cases

Sample Input

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```
2
2 1
1 2
2 1
1
4 2
4 2 1 3
1 3 4 2
2
3
```

Sample Output

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```
2
3
3
```

Sample Explanations

For sample case 1, there are 2 players and 1 query, the initial starting position of player 1 is position 1 and the initial starting position of player 2 is position 2.

The shuffle permutation denotes that every time we shuffle, the player in position 1 moves to position 2 and the player in position 2 moves to position 1.

To answer the query, we shuffle once, getting the starting positions 2 1. In the first and only round of the tournament, Player 1 will play against Player 2. Since Player 2 has the higher power, Player 2 wins, and the output is 2.

For sample case 2, we have 4 players and 2 queries. The initial starting positions of the players has Player 4 in position 1, Player 2 in position 2, Player 1 in position 3, and Player 3 in position 4.

Every time we shuffle, the player in position 1 moves to position 2, the player in position 2 moves to position 3, the player in position 3 moves to position 4, and the player in position 4 moves to position 1.

For the first query of 2, the starting positions become 4 2 1 3 -> 4 3 2 1 -> 4 1 3 2 after two shuffles.

In the first round of the tournament, Player 4 plays Player 1, wins, and now has a power of 1. Player 3 plays Player 2, wins, and now has a power of 2. Then Player 3 plays Player 4 in the finals and since Player 3 has more power, they win the tournament and so the output to the query is 3.

For the second query of 3, this means we need to apply 3 more shuffles to the positions after our previous query, 4 1 3 2 -> 4 2 1 3 -> 4 3 2 1 -> 4 1 3 2. After shuffling, the winner of the new tournament for the second query is again player 3.